

Joe Barlow

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SUMMARY

Backend engineer with 5+ years in a Django monolith serving thousands of small businesses — API design, endpoint and query performance, and async processing across 512 PostgreSQL shards — and four years before that in customer support at the same company, talking to those businesses every day. That path shapes how I build: I start from what the user is trying to do and work backward to the architecture. Equally at home evolving critical systems that already carry load and building new ones from zero.

TECHNICAL SKILLS

Languages & Frameworks: Python, Django, Django REST Framework, SQL, JavaScript

Data: PostgreSQL (sharded), BigQuery, Redis, data modeling, SQL optimization, ETL pipeline design

Infrastructure: Google Cloud Platform (GKE, GCS, BigQuery), Kubernetes, Docker, Celery, CI/CD, observability & alerting

Practices: REST API design, zero-downtime migrations, caching, async/event-driven architecture, LLM developer tooling

EXPERIENCE

Senior Software Engineer · Shopventory · Feb 2025 – May 2026 · Remote

- Own backend architecture, reliability, and production health for a multi-tenant Django platform serving thousands of SMBs — API design, async task processing, and database operations across 512 PostgreSQL shards.
- Led a zero-downtime, two-year data migration across every shard in the fleet, sequencing schema changes and incremental rollouts so critical operations kept running. No data loss, no customer-visible incidents.
- Built our cloud data warehouse analytics platform from zero — chose the architecture, data model, ETL pipelines, wrote query patterns, then stood up migration sequence and owned it in production. Cut report load times up to 100x on high-volume datasets, and designed the API surface with cross-team collaboration.
- Serve as the team's go-to reviewer and sounding board for earlier-career engineers — architectural direction on new work, code review, and working through problems live when someone gets stuck. Lead team meetings and present technical work internally.
- Built LLM-powered developer tooling now used daily across the team: custom Claude Code commands that encode our working principles into the plan/execute loop, a code review bot, and a CI step that flags end-of-life Python dependencies and opens upgrade PRs. Measurably faster onboarding and day-to-day codebase navigation.
- Drove major platform upgrades (Python, Django, PostgreSQL, CI/CD), choosing incremental rollouts over rewrites.
- Partner directly with the CTO and founding engineers on architecture, roadmap, and cross-team technical decisions.

Software Engineer · Shopventory · Apr 2021 – Feb 2025 · Remote

- Designed and shipped REST APIs and backend services powering inventory, purchasing, and reporting, plus integrations with Stripe, Square, and Shopify — unifying workflows that small businesses previously ran by hand.
- Delivered features end-to-end alongside product, from shaping requirements through API design, implementation, and rollout — including frontend work where needed.
- Built high-volume async data flows between internal systems and third-party APIs, tuned for reliability under sustained load.
- Improved system reliability through monitoring, alerting, and performance tuning across Kubernetes infrastructure.

Customer Care Lead / Representative · Shopventory · Feb 2017 – Apr 2021 · Denver

- Spent four years on the phone with small business owners — thousands of conversations about what was broken, confusing, or costing them money. Led the support team before moving into engineering at the same company.
- That work is why I read a spec by asking who is on the other end of it, and still debug from the customer's side of the system first.

PROJECTS

Flock Manager · Solo Developer · iOS & Android 2025 – Present

- Took a cross-platform mobile app for backyard flock management from empty repo to both the Apple App Store and Google Play, solo.
- Owned every decision: data modeling, system design, UX, beta recruitment, store submissions, and post-launch support.
- Stack: Flutter, Dart, SQLite · github.com/pingponghero/flockmanager

EDUCATION

Grove City College — B.S.E.E., Electrical Engineering

Denver Seminary — Master of Divinity